

Vatsal Verma

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience engineering event-driven microservices and backend infrastructure for enterprise billing platforms serving 80M+ subscribers. Specialized in building resilient systems using Java, Spring Boot, Apache Kafka, Couchbase, and PostgreSQL, with containerized deployments across Kubernetes and OpenShift. Strong focus on secure RESTful API design, performance optimization, and platform modernization, highlighted by shipping a production Agentic RAG system that cut root cause analysis time by 20%.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend & Distributed Systems: Spring Boot, Spring Cloud, Spring Security, microservices, REST API design, OpenAPI, OAuth2/JWT, API rate limiting, concurrency and multithreading, event-driven architecture

Data, Search & Messaging: Apache Kafka, Couchbase, PostgreSQL, Elasticsearch, caching

AI / GenAI: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI agents, LangGraph, Google Agent Development Kit (ADK), Model Context Protocol (MCP), vector databases, embeddings, semantic search, hybrid retrieval (BM25, kNN)

DevOps & Testing: Docker, Kubernetes, Red Hat OpenShift, Linux, Jenkins, Maven, Git, CI/CD, JUnit, TestNG, Mockito, SonarQube

PROFESSIONAL EXPERIENCE

AI Software Engineer (*SWE-2*)

August 2023 – Present

Amdocs, Pune, India

- Owned end-to-end delivery of business-critical features within a large-scale billing ecosystem comprising 36 microservices for Tier-1 telecom clients including AT&T and Bell Canada, collectively serving 80M+ subscribers.
- Engineered event-driven microservices and RESTful APIs in Java, Spring Boot, and Spring Cloud, processing 50M+ billing events per day across 40+ Kafka topics with Couchbase sustaining 50K+ ops/sec, specified in OpenAPI and secured with OAuth2/JWT authentication and per-client rate limiting.
- Architected a production-grade Agentic AI-powered automated incident triage solution with LangGraph and Google ADK, leveraging a complete RAG pipeline with hybrid retrieval (BM25 and dense vector search) and Elasticsearch (kNN, dense_vector) as the knowledge repository for embeddings and historical root cause analyses.
- Integrated real-time Apache Kafka event streams, RESTful APIs, and MCP (Model Context Protocol) tool interfaces to enrich AI agent context, enabling engineering teams to rapidly identify, diagnose, and reprocess exception records, ultimately reducing root cause analysis time by 20% and significantly decreasing manual incident handling efforts.
- Streamlined Maven-based CI/CD pipelines on Red Hat OpenShift by distributing TestNG suites across Kubernetes API-provisioned clusters, cutting integration test execution time by 25%.
- Managed production release cycles and cleared the pre-release security gate by remediating 42 CVEs and OWASP Top 10 findings in application dependencies and container base images.

AI/ML Research Intern

August 2021 – February 2022

Indian Institute of Technology (IIT-BHU), Varanasi, India

- Built facial recognition models using Convolutional Neural Networks (CNNs), Histogram of Oriented Gradients (HOG), TensorFlow, and Keras to support identification and classification of 50,000+ individuals.
- Applied computer vision techniques to agricultural research by analyzing unique muzzle patterns of over 100 cattle, extending facial recognition methodologies to livestock identification use cases.
- Investigated deep learning algorithms and image processing pipelines while effectively managing research timelines and presenting technical findings, analyses, and recommendations to faculty advisors.

PROJECTS

Access Control for Multi-Agent MCP Servers

Java 21, Spring Boot, Spring Security, Maven, MCP

- Built an open-source Java 21 library adding profile-based RBAC to multi-agent MCP servers, tagging tools, resources, and prompts with profiles to enforce least-privilege capability discovery and fail-closed execution per agent.
- Architected a five-module Maven codebase with Spring Boot auto-configuration, JWT and hashed API-key authentication via Spring Security, signed session tokens narrowing permissions across agent delegation, and a CLI for policy validation.

AI-Powered Animal Detection and Alert System

Python, TensorFlow, Keras, OpenCV, REST API

- Developed an end-to-end deep learning inference pipeline utilizing CNNs models, TensorFlow, Keras, and OpenCV to enable real-time animal detection with an accuracy rate of 92.68%.
- Implemented image preprocessing and data augmentation techniques to improve model robustness and ensure reliable performance across varying environmental and lighting conditions.
- Integrated low-latency event notification capabilities using the Twilio REST API to deliver real-time alerts, earning recognition at WildHacks hackathon for innovative and creative API utilization.

EDUCATION

Vellore Institute of Technology, Vellore

July 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering with specialization in IoT

CGPA: 8.22/10

ACHIEVEMENTS AND LEADERSHIP

- Earned the Innovation Award at Amdocs** for enhancing TestNG suite execution efficiency and optimizing CI/CD processes through Kubernetes API-based cloud cluster integration.
- Secured recognition at WildHacks 2021** for delivering the most creative implementation of the Twilio API during a competitive hackathon.
- Led the design team for the ISA-VIT Student Chapter** by managing a team of 15 members and overseeing the design, development, and successful release of an application on the Google Play Store.